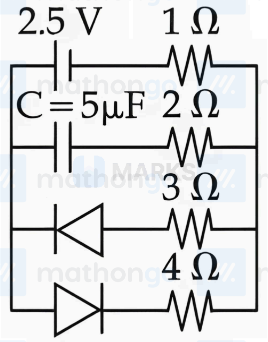
Q1. JEE Main 2026 (21 January Shift 2)

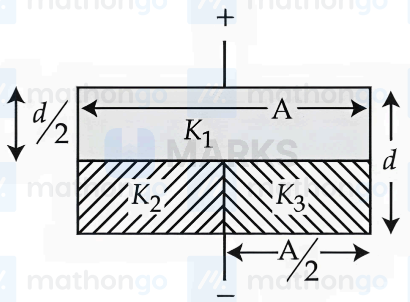
The charge stored by the capacitor C in the given circuit in the steady state is ____ μC.



(1) 5 (2) 12.5 (3) 10 (4) 7.5

Q2. JEE Main 2026 (23 January Shift 1)

The space between the plates of a parallel plate capacitor of capacitance C (without any dielectric) is now filled with three dielectric slabs of dielectric constants $K_1 = 2, K_2 = 3$ and $K_3 = 5$ (as shown in figure). If new capacitance is $\frac{n}{3}C$ then the value of n is ____.



Q3. JEE Main 2026 (28 January Shift 2)

Identify the correct statements :

A. Effective capacitance of a series combination of capacitors is always smaller than the smallest capacitance of the capacitor in the combination.

B. When a dielectric medium is placed between the charged plates of a capacitor, displacement of charges cannot occur due to insulation property of dielectric.

C. Increasing of area of capacitor plate or decreasing of thickness of dielectric is an alternate method to increase the capacitance.

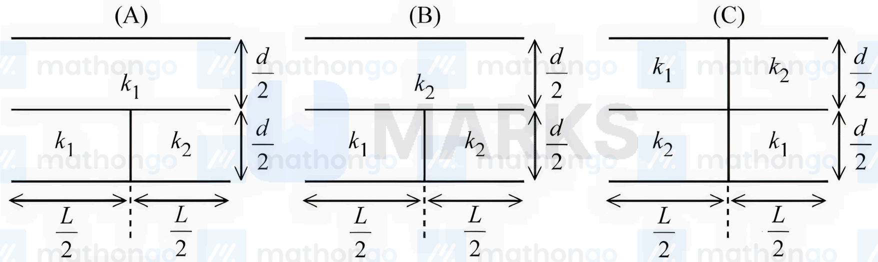
D. For a point charge, concentric spherical shells centered at the location of the charge are equipotential surfaces.

Choose the correct answer from the options given below :

(1) B and D Only
(2) A, B and C Only
(3) C and D Only
(4) A, C and D Only

Q4. JEE Main 2026 (24 January Shift 2)

Three parallel plate capacitors each with area A and separation d are filled with two dielectric (k_1 and k_2) in the following fashion. Which of the following is true? ($k_1 > k_2$)



(1) $C_C > C_A > C_B$
(2) $C_C > C_B > C_A$
(3) $C_A > C_C > C_B$
(4) $C_B > C_C > C_A$

Q5. JEE Main 2026 (22 January Shift 2)

A capacitor P with capacitance 10×10^{-6} F is fully charged with a potential difference of 6.0 V and disconnected from the battery. The charged capacitor P is connected across another capacitor Q with capacitance 20×10^{-6} F. The charge on capacitor Q when equilibrium is established will be $\alpha \times 10^{-5} C$ (assume capacitor Q does not have any charge initially), the value of α is ____。

Q6. JEE Main 2026 (23 January Shift 2)

A parallel plate capacitor with plate separation 5 mm is charged by a battery. On introducing a mica sheet of 2 mm and maintaining the connections of the plates with the terminals of the battery, it is found that it draws 25% more charge from the battery. The dielectric constant of mica is ____.

(1) 1.0
(2) 2.5
(3) 2.0
(4) 1.5

Q7. JEE Main 2026 (21 January Shift 1)

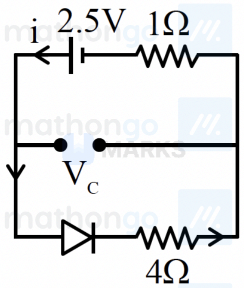
A parallel plate capacitor has capacitance C, when there is vacuum within the parallel plates. A sheet having thickness $\left(\frac{1}{3}\right)^{\text{rd}}$ of the separation between the plates and relative permittivity K is introduced between the plates. The new capacitance of the system is :

(1) $\frac{4KC}{3K-1}$
(2) $\frac{3CK^2}{(2K+1)^2}$
(3) $\frac{3KC}{2K+1}$
(4) $\frac{CK}{2+K}$

ANSWERS AND SOLUTIONS

1. (3) **2.** 8 **3.** (4) **4.** (3) **5.** 4 **6.** (3) **7.** (3)

1. **(3)** In steady state:



$i = 2.5/5 = 0.5$ A

$V_c = 4 \times 0.5$

$V_c = 2V$

charge

$Q = CV_c$

$= 5 \times 2$

$= 10\,\mu C$

2. **(8)** The initial capacitance is $C = \frac{\epsilon_0 A}{d}$.

From the figure, the capacitor is divided into three parts.

The bottom half is divided into two capacitors C_2 and C_3 in parallel, each with area $A/2$ and thickness $d/2$.

$C_2 = \frac{K_2\epsilon_0(A/2)}{d/2} = K_2\frac{\epsilon_0 A}{d} = 3C$

$C_3 = \frac{K_3\epsilon_0(A/2)}{d/2} = K_3\frac{\epsilon_0 A}{d} = 5C$

The equivalent capacitance of the bottom part is $C_{eq,bottom} = C_2 + C_3 = 3C + 5C = 8C$.

The top half is a single capacitor C_1 with area A and thickness $d/2$.

$C_1 = \frac{K_1\epsilon_0 A}{d/2} = 2K_1\frac{\epsilon_0 A}{d} = 2(2)C = 4C$.

The total capacitance C' is the series combination of C_1 and $C_{eq,bottom}$.

$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_{eq,bottom}} = \frac{1}{4C} + \frac{1}{8C} = \frac{2+1}{8C} = \frac{3}{8C}$

$C' = \frac{8}{3}C$

Comparing with $C' = \frac{n}{3}C$, we get $n = 8$.

3. **(4)** A. Effective capacitance of series combination is always smaller than the smallest capacitance - TRUE. For series: $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$, therefore $C_{eff} < C_{min}$

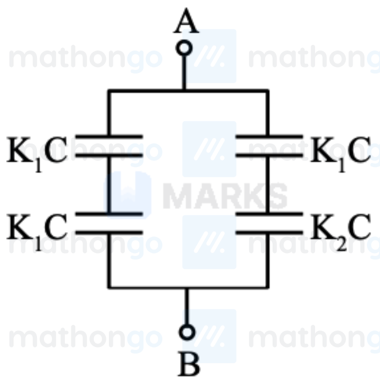
B. Dielectric prevents charge displacement - FALSE. A dielectric allows polarization (displacement of bound charges) but prevents free charge movement. It does not prevent displacement of bound charges.

C. Increasing area or decreasing thickness increases capacitance - TRUE. From $C = \frac{\varepsilon_0\varepsilon_r A}{d}$, increasing A or decreasing d increases C.

D. Concentric spherical shells centered at a point charge location are equipotential surfaces - TRUE. For a point charge, the potential at distance r is $V = \frac{kq}{r}$, which depends only on distance, making concentric spheres equipotential surfaces.

Correct statements: A, C, D

4. **(3)** For C_A

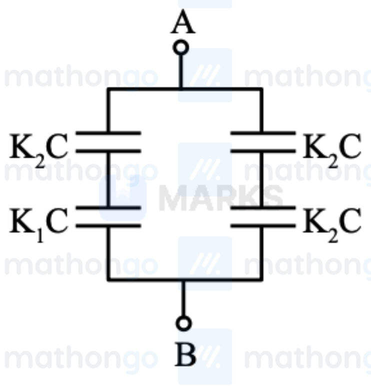


Let $\frac{\varepsilon_0 A}{d} = C$

$\therefore C_A = \frac{K_1 C}{2} + \frac{K_1 K_2 C}{K_1 + K_2}$

$= K_1 C \left[\frac{K_1 + 2K_2}{2(K_1 + K_2)}\right]$

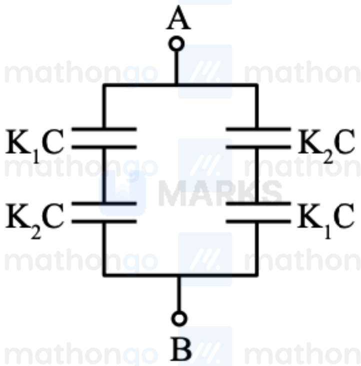
For C_B



$C_B = \frac{K_2 C}{2} + \frac{K_1 K_2 C}{K_1 + K_2}$

$= K_2 C \left[\frac{K_1 + 2K_2}{2(K_1 + K_2)}\right]$

For C_C



$\therefore C_C = \dfrac{2K_1K_2C}{K_1+K_2}$

$C_A > C_C > C_B$

5. **(4)** Capacitor P (10×10^{-6} F) is charged to 6.0 V.

Initial charge: $Q_P = 10 \times 10^{-6} \times 6 = 60 \times 10^{-6}$ C.

When connected to uncharged capacitor Q (20×10^{-6} F), charge redistributes until both reach same potential.

Final voltage: $V_f = \frac{Q_{total}}{C_{total}} = \frac{60\times10^{-6}}{30\times10^{-6}} = 2$ V.

Charge on Q: $Q_Q = 20 \times 10^{-6} \times 2 = 40 \times 10^{-6}$ C = 4×10^{-5} C.

Thus $\alpha = 4$.

6. **(3)** Initial capacitor has plate separation 5 mm with capacitance $C_1 = \epsilon_0 A/5$.

After inserting 2 mm mica sheet, the gap remaining is 3 mm of air. The effective separation becomes: $d_{eff} = 3 + 2/k$ where k is the dielectric constant of mica.

New capacitance: $C_2 = \frac{\epsilon_0 A}{3+2/k}$

Since battery remains connected, voltage is constant. The charge is: $Q = CV$

Ratio of charges: $\frac{Q_2}{Q_1} = \frac{C_2}{C_1} = \frac{\epsilon_0 A/(3+2/k)}{\epsilon_0 A/5} = \frac{5}{3+2/k}$

Given that charge increases by 25%: $Q_2 = 1.25Q_1$

$\frac{5}{3+2/k} = 1.25$

$5 = 1.25(3 + 2/k)$

$5 = 3.75 + 2.5/k$

$1.25 = 2.5/k$

$k = 2$

7. **(3)** Original capacitance: $C = \frac{\varepsilon_0 A}{d}$

After inserting dielectric of thickness $\frac{d}{3}$, system becomes two capacitors in series.

Vacuum region: $C_1 = \frac{\varepsilon_0 A}{2d/3} = \frac{3C}{2}$

Dielectric region: $C_2 = \frac{K\varepsilon_0 A}{d/3} = 3KC$

Equivalent capacitance: $\frac{1}{C_{new}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{2}{3C} + \frac{1}{3KC} = \frac{2K+1}{3KC}$

$C_{new} = \frac{3KC}{2K+1}$